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Sustainable Education

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- **Key methods in ESD**
- 1. Collaborative real-world projects such as a service-learning project and campaigns for different sustainability topics;
- 2. Vision-building exercises such as future workshops, scenario analyses , utopian/dystopian story-telling, science-fiction thinking, and fore and back-casting;
- 3.Analysis of complex systems including community-based research projects, case studies, stakeholder analysis, actor analysis, modelling and systems games;
- 4. Critical and reflective thinking including through fish-bowl discussions and reflective journals.
- Education plays a synergistic role in achieving the aspirations embedded within Agenda 2030 and the SDGs. Thus, we may conclude that Education for Sustainable Development (ESD) will require individuals to acquire ‘key competencies’, aligning with notions of transformational learning, in addition to other generic and context specific competencies.

Table 2. Competencies proposed for assessment of ESD, including addressing Sustainable Development Goal (SDG) 4.7 (informed by Wiek et al. [13]) plus evaluation questions.

Key Competency Area	Example of Competencies	Example Evaluation Questions
Intrapersonal	Presencing, self awareness, stress management, meaning making, connection with self, capacity for inner peace, mental wellbeing, self-reflection	Are learners able to be present in themselves? Can learners hold (without having to resolve them or prejudice one or the other) contradictory feelings and or thoughts? Do learners practise self awareness? Are learners able to know when they or a group is stressed and take appropriate steps so that stress does not hinder action? Can learners find strategies to seek inner peace? Can learners make meaning in the work they do? Do learners practise love and compassion? Are learners aware of their mental and emotional health and do they have the abilities to maintain healthy mental and emotional states?
Interpersonal	Communication skills, empathy, compassion, leadership, teamwork, mediation, cooperation, collaboration, participation	Are communication skills taught? Are learners facilitated to work well with others? Can learners assist each other in peer to peer learning? Are learners, across gender, ethnicity and other groupings, able to explore their leadership skills? Is empathy valued and encouraged? Are learners able to address conflict and develop mediation skills? Are their barriers to full participation in learning projects?
Future thinking	Visioning, developing scenarios, backcasting, recognising heritage, intergenerational equity	Are learners encouraged to imagine and envision sustainable futures? Can learners effectively use backcasting and forecasting skills in planning strategic activities? Do learners connect with their heritage and culture when looking to the future? Can learners identify future scenarios and use them to inform decision making? Are learners able to apply an awareness of intergenerational fairness to decisions and planning?
Systems thinking	Systems thinking, working with complex problems, promoting resilience, understanding tipping points and feedback loops	Are learners able to work with interconnectedness and complexity in a systemic context? Do learners have a functional knowledge of tipping points, resilience and feedback loops? Can learners understand how to work with socio-ecological systems? Do learners have a working concept of resilience?
Disciplinary and interdisciplinary	Understand the links between knowledge and experience, critical thinking, discipline specific framing, interdisciplinarity, expressing multiple ways of knowing	Have learners acquired an epistemological intelligence? Have learners developed awareness of different ways of knowing? Have learners explored disciplinary integrity and understood the academic norms of a discipline? Can learners work with disciplines that are not their core approach? Have learners developed their capacities for critical thinking? Can learners critically reflect on their own experiences?
Normative and cultural	Ethical responsibility, development of world views and perspectives, awareness of values, understanding of justice, cosmopolitan perception, transcultural understanding, awareness of local context and global trends	Can learners identify ethical questions and evaluate ethical responses according to different frameworks? Are fairness and justice debated and explored? Are learners encouraged to engage with and understand different world views? Are different cultural contexts appreciated? Have learners engaged with questions of well being and happiness?
Strategic	Planning, decision making, implementing, addressing challenges, organisational development, use of Kolb's action reflection cycle.	Are learners able to practise decision making and analyse consequences? Can learners use planning and assessment tools? Can learners identify and address challenges with regard to strategies and their implementation? Have learners implemented a plan they have designed? Do learners know how to use the behavioural change cycle for effective action and reflection? Are learners aware of organisational development issues and practices?

Haan et al. and Barth et al. have also suggested for the following **intrapersonal competencies** :

1. Presencing : The ability to stay present to your internal environment at the same time as engaging with your external environment.
2. The ability to hold contradictory thoughts and feelings without having to resolve the contradictions.
3. Knowledge of stress and how to know when you are stressed and what can help you to reduce your stress and avoid burnout.
4. The ability to cultivate awareness; the skill to be present and out of that presence become aware of states of being beyond your rational mind.
5. The knowledge and ability to find inner states of peace and compassion, for one's self and others.
6. The ability to make meaning out of experience; and the ability to synthesize experience, models or frameworks, and feed back into previously unknown meta-perspectives.
7. The ability to experience and deepen love and connection to yourself, other humans, and the non-human world.

- Using the Competencies and Assessment Framework Against SDG 4.7(ESD & Global Citizenship)
- Human rights
- Gender equality
- Promotion of a culture of peace and non-violence
- Global citizenship
- Appreciation of cultural diversity and of culture's contribution to sustainable development
- Activating Competencies through Multiple Intelligences
- The multiple intelligences proposed by Howard Gardner , represent a sum total of human capacities to influence, interact with, and communicate with our world. This includes both human and non-human life forms. The seven intelligences are represented in the Figure given here.

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Figure 1. The multiple intelligence model developed by Gardner [16]—image http://thedailynewnation.com.

Table 3. Key competencies with reference to multiple intelligences, applied to the areas indicated in SDG4.7. These particular questions should be used together with the questions in Table 2.

Key Competencies	Human Rights	Gender Equality	Culture of peace and non Violence	Global Citizenship	Cultural Diversity
Interpersonal. Collaborative skills, mediation, leadership, cooperation, empathy, teamwork	Are learners given the opportunity to develop empathy? What is leadership in promotion of human rights? What role does cooperation play in human rights?	Do learners have the possibility of experiencing their world from the viewpoint of the opposite sex? Do opposite sexes have opportunities to collaborate together on shared goals? Do both sexes look together at issues of gender equality in a spirit of enquiry? Are different forms of gender included and enabled?	Are there opportunities to explore peace or non-violence between groups or individuals in everyday contexts? Are any peace 'technologies' such as non-violent communication taught?	Can learners explore what it means to be a member of a particular social or ethnic group and a citizen of the world? How might that lead to differing actions or ways of thinking or being? Are learners given the opportunity to learn the international mechanisms of global cooperation?	Do learners have the opportunity to reflect and understand their culture's attitudes to others and 'otherness'? In what way can local cultural activities enable empathy and appreciation of cultural diversity?
Strategic Planning. Decision making strategies, awareness of success factors, obstacles to change, knowledge of behavioural change, organisational development skills	Do learners know what successful strategies for enjoyment of human rights have been employed in their countries? What is in the way of greater employment of human rights in their country? Can learners identify the changes in individual, group, or national behaviours that are most likely to lead to more human rights being enjoyed by more people?	Can learners find strategies that will lead to greater gender equality? What might be the obstacles to greater gender equality? Can learners identify what changes in organisations would lead to greater gender equality?	Can learners identify strategies for peace and non-violence? Can learners identify the barriers to peace at any levels of scale? Do learners experience differing organisational structures and what their role promoting peace and non violence might be like?	What strategies can lead to great engagement with global citizenship amongst their culture or country? What strategies might be employed to grow global citizenship awareness and action as opposed to nationalistic behaviour and attitudes?	Can learners express using cultural avenues ways to change behaviour or make change happen?
Normative Competencies. Knowledge of the sustainability of current or future states, knowledge of and awareness of Justice, fairness, happiness, well being, risk, trade offs, and ethical questions	Do learners explore where human rights come from and how? Can learners identify which human rights most directly affects personal happiness or well being? Do learners have the possibility of reflecting on the current level of human rights in their culture or country? Do learner reflect on the trade offs between different human rights and their consequences?	Can learners identify rightness or wrongness of the current state of gender equality? Do all genders have the possibility to fulfil their potential in education? Do learners identify In what way does gender determine levels of happiness or well being?	Can learners reflect on the relative sustainability of cultures of violence? Learners can reflect on how violence affects the sustainability of that society. Learners can reflect on the trade offs between violence and peace in their society. Learners can reflect on what risk a culture of peace carries.	Do learners have the opportunity to reflect on the risks and rewards of adopting global perspectives to themselves or their society? Can learners reflect on to what extent global citizenship is encouraged, or discouraged in their society? Do learners examine what improved global citizenship awareness might have in a an imagined future?	Can learners see the role their culture norms and values plays in promoting happiness, well being, justice, or fairness? Do learners reflect on how their culture engages with ethical questions and issues particularly around diversity? How are levels of diversity in their culture affecting general levels of well being? Can learners assess any risks or trade offs in their cultural diversity?
Anticipatory Skills. Working with scenarios, forecasting and backcasting, intergenerational equity	Can learners agree on a date in time where full implementation of Human rights be a reality and backcast the necessary steps until today, considering both incremental and transformative steps? Can learners imagine different scenarios or pathways of achieving plenum human rights in their country?	Have learners examined how inter generational equity has affected gender equality?	Can learners foresee a time in their country where a culture of peace and non-violence is a reality? Can they backcast these scenarios? Are they able to imagine or design different pathways with incremental or transformational steps? Do learners understand or know the process of intergenerational culture reproductions (memes) and ways of transforming them?	Can learners foresee a time when global citizenship has achieved equality with national citizenship	Can learners anticipate and outline pathways to a culture of equality of diversity? Can learners imagine or plot pathways to when questions of diversity become irrelevant?

contd…

Table 3. *Cont.*

Key Competencies	Human Rights	Gender Equality	Culture of peace and non Violence	Global Citizenship	Cultural Diversity
System Thinking. Ability to work with Feedback loops, systems and sub-systems, buffers and multiple variables, nested scales, resilience, and tipping points.	Do learners have opportunities to reflect on the role human rights play in changing systems of power or oppression? Can learners identify how human rights feature in human social or economic systems?	Are learners able to reflect on how social political and economic systems are distorted by gender inequality? What feedback loops might be operating in issues of gender in their culture?	Can learners build or model social or political systems of peace? What are the feedback loops (social, economic, political, or ecological) which create or maintain violence? Can learners reflect on levels of 'social capital' that need to be maintained to ensure a peaceful society?	How can system change knowledge be used to increase or decrease knowledge of or ability to act as a global citizen?	How do diversity issues influence tipping points in social, environmental, or cultural change? Is changing the diversity of a culture a key step towards initiating cultural change?
Intrapersonal Competency. Presencing Ability to hold contradictory feelings and thoughts Personal and group stress management Cultivating awareness Finding inner peace and compassion, meaning making, Experiencing Love and connection	Can learners reflect on where the impulse for humans rights spring? What internal awareness or competencies enhance or detracts from the societal recognition of human rights? Can learners reflect on the effect human rights or the lack of human rights has on personal feelings of safety or peace? Can learners reflect on the role of love and compassion on human rights?	Can learners reflect on how the present level of gender equality affects their inner states of safety, compassion, stress, and connection? To what extent are cultural levels of stress affected by gender inequality?	Can learners reflect on peace and what levels of peace they find in themselves? Can learners reflect on where violence comes from in themselves and what makes a violent response more or less likely? Can learners hold or be present to violence with equanimity? Can learners reflect on the effect of a non-violent approach to communication has on their inner states?	Can learners reflect on what it feels like to them being both a person of a place and a global citizen? Can learners reflect on how an awareness of a global perspective changes their sense of themselves? Can learners reflect on how awareness global citizenship increases or decreases their levels safety or well being?	Can learners reflect on how their cultural values related to diversity increase or decrease their feels of presence? Can learners reflect on what effect cultural expressions like dance or singing has in their inner states? Can learners reflect on how cultural expressions like art or music can change their personal experience of being in a group?

ESD Assessment as Part of an Active Learning Cycle

- The framework is not linear and is intended to be part of an iterative action/reflection cycle of planning, implementing, learning, reflection and re-planning. This framework recognizes the positive learning cycle in monitoring and evaluation in which learning stimulates outcomes which are evaluated, feeding into reflection and planning for future implementation of teaching.

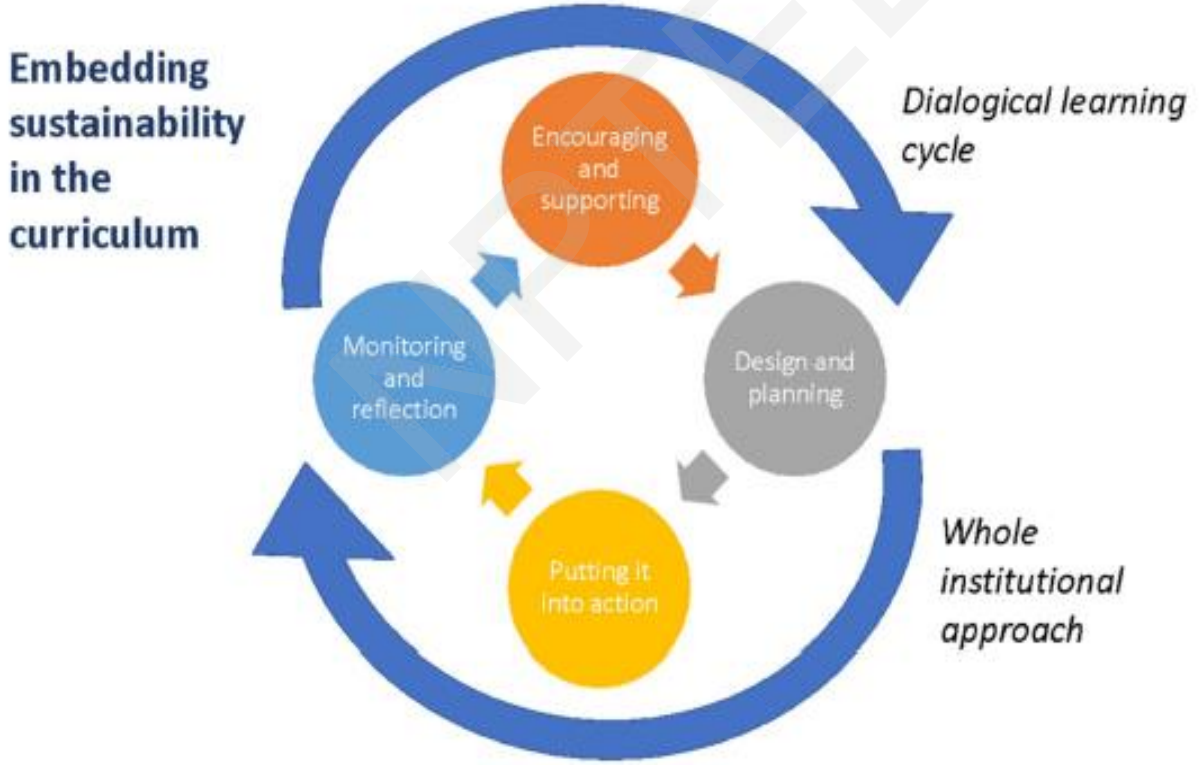
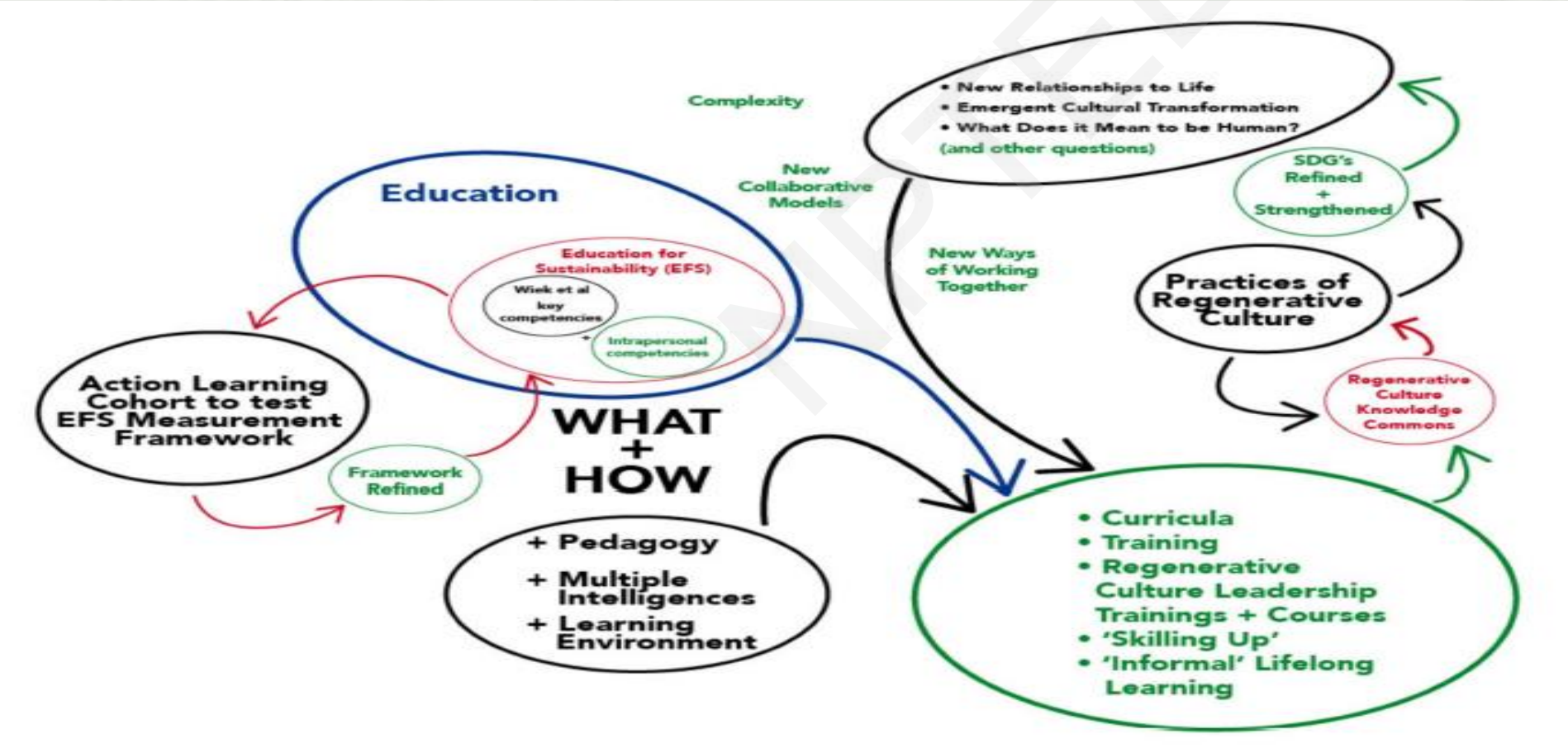


Figure 2. The action–reflection cycle of learning and assessment: Drawing on Kolb and Kolb [

- An illustration of how monitoring and assessment of ESD through competencies forms part of an active learning cycle, which together with pedagogy and learning environment integrate with curricula for formal and non-formal education. Education in this sense forms part of our wider practices for regenerative culture.



We are increasingly asking if what people learn is truly relevant to their lives, if what they learn helps to ensure the survival of our planet. Education for Sustainable Development can provide the knowledge, awareness and action that empower people to transform themselves and transform societies.

- Stefania Giannini, Assistant Director-General for Education, UNESCO

Societal transformation:
Enable the achievement of the SDGs towards building a more sustainable world

Pedagogy and learning environment:
Employ interactive, project-based, learner-centred pedagogy. Transform all aspects of learning environment through a whole-institution approach to ESD to enable learners to live what they learn and learn what they live

Learning outcomes:
Empower people to take responsibility for present and future generations and actively contribute to societal transformation

Learning content:
Integrate sustainability issues, in particular those enshrined in the 17 SDGs such as climate change, into all kinds of learning

The General Assembly reaffirms education for sustainable development as an integral element of the Sustainable Development Goal on quality education and a key enabler of all the other Sustainable Development Goals.

UN General Assembly Resolution 72/222 (2017)

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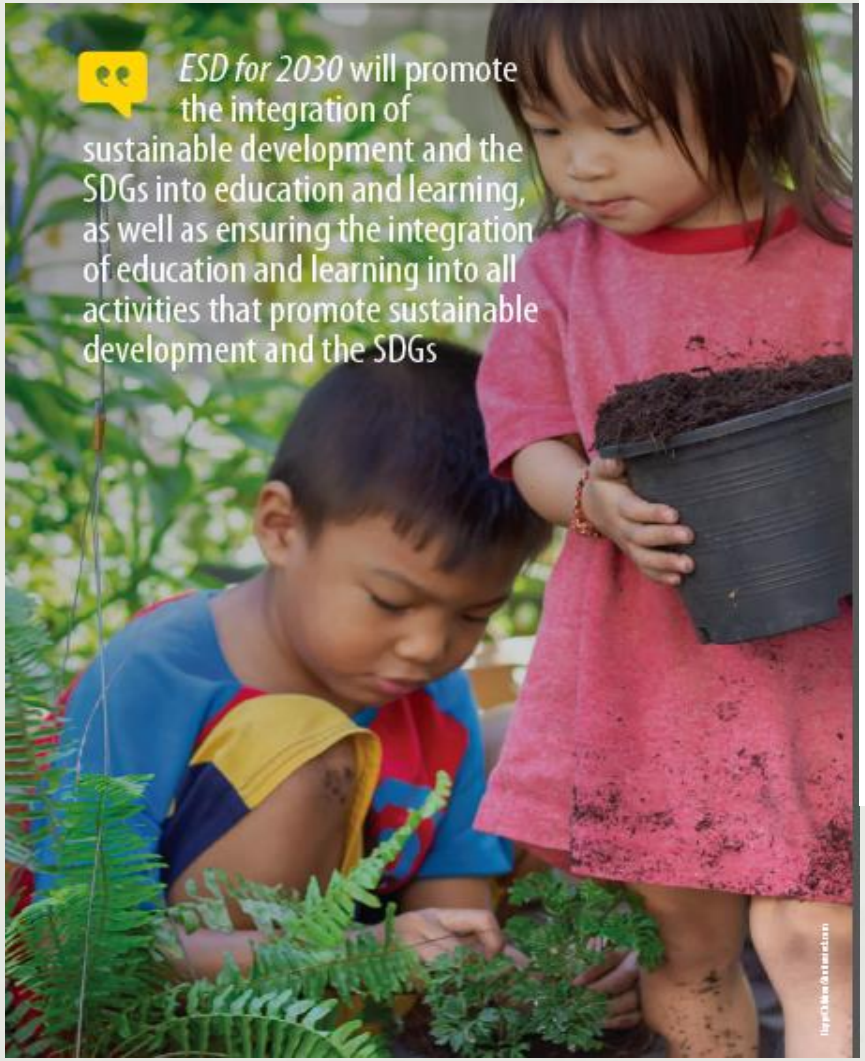

Education for Sustainable Development aims to raise knowledge, awareness and action

Focus on the big Transformation

- ESD must focus on the big transformation that is needed for sustainable development and provide relevant educational interventions. The idea of big transformation implies changes in individual action intertwined with reorganization of societal structures, and it requires ESD to track the transformation towards a more just and sustainable direction.
- Transformation necessitates, among other things, a certain level of disruption, with people opting to step outside the safety of the status quo or the "usual" way of thinking, behaving or living. It requires courage, persistence and determination, which can be present at different degrees, and which are best sourced from personal conviction, insight, or the simple feeling of what is right.

What are the key reflections that inform ESD for 2030?

✓ **Transformative action**: Fundamental changes required for a sustainable future start with individuals. ESD has to place emphasis on how each learner undertakes transformative actions for sustainability, including the importance of opportunities to expose learners to reality, and how they influence societal transformation towards a sustainable future. ESD in action is citizenship in action.

✓ **Structural changes**: ESD must pay attention to the deep structural causes of unsustainable development. Balancing act between economic growth and sustainable development is needed and ESD should encourage learners to explore alternative values to those of consumer societies, as well as having a structural view on how to address ESD in the context of extreme poverty and vulnerable situations.

✓ **The technological future**: ESD has to respond to the opportunities and challenges brought about by technological advances. Some 'old' problems will be resolved through technology, but new challenges and risks will arise. Critical thinking and sustainability values become ever more relevant, as the task of teaching ESD may become more challenging with the illusion that technologies can resolve the majority of sustainability problems.

Transformative Learning: A Paradigm Shift for Sustainable Education in India

- Transformative learning involves experiencing a deep, structural shift in the basic premises of thought, feelings, and actions. It is a shift of consciousness that dramatically and irreversibly alters our way of being in the world. Such a shift involves our understanding of ourselves and our self-locations; our relationships with other humans and with the natural world; our understanding of relations of power in interlocking structures of class and gender; our body awareness , our visions of alternative approaches to living; and our sense of possibilities for social justice and peace and personal joy (O'Sullivan (2003).
- Transformative Perspective of Learning as Paradigm shift (Mezirow, 1997)

'The process of becoming critically aware of how and why our assumptions have come to constrain the way we perceive, understand and feel about our world…changing these structures of habitual expectation make possible a more inclusive, discriminating and integrating perspective…and involve making choices or otherwise acting upon these new understandings'.

- **Transformative Learning**
- It started in 1978 and is defined as an orientation which holds the way learners interpret their sense experience is central to transformative learning.
- This kind of learning experience involves a fundamental change in our perspectives
- Learners started to question all the things they knew to make room for new insights and information.
- Transformative learning has two major focuses: Instrumental learning and Communicative learning.

- **The perspective transformation is explained as follows:**
- Disorienting Dilemma
- Self examination
- Sense of alienation
- Relating discontent to others
- Explaining options of new behavior
- Building confidence in new ways
- Transformative Education
- The three main pillars of transformative learning are:-
- 1. Analysis of current situation
- 2. Vision to look like an alternative to the prevailing model.
- 3. The process of change towards responsible action

- Transformative education provides for decision making process in all these phases
- The goal is to enable students to understand the problem and at the same time empower them with the knowledge to counter the challenges
- **The phases of Transformative Learning-**
- **A disorienting dilemma**: When the learner finds what they believed in the past is not accurate
- **Self-examination:** Students will do a self examination of their beliefs and understanding and connect this to disorienting dilemma
- **Critical Assessment of assumptions**: Students are able to take a comprehensive view at their past beliefs and review them critically
- **Planning a course of action**: when the students realize that their past understanding was faulty , they are able to plan a course of action
- **Acquisition of knowledge to carry out new plans**: This is the time to carry out their plan and get ahead in the way of transformative learning
- **Exploring and trying new roles**: In transformative learning , exploring and trying new roles is a key to success.

Mezirow's 10 Phases

1. A disorienting dilemma
2. A self examination
3. A critical assessment
4. Recognition
5. Exploration
6. Planning of a course of action
7. Acquisition of knowledge
8. Provisional trying of roles
9. Building of competence self confidence
10. A reintegration

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- **Principles of Andragogy-**
- There are six assumption that are often stated as principles of adult education:
- Need to know why
- Self-Concept
- Role of experience
- Readiness to learn
- Orientation to learning
- Motivation

- **Analysis on how to foster Transformative Learning:**
- Some factors that can foster transformative learning in online environment are:-
- Relationship: Transformative learning can be fostered by establishing supportive and trusting relationship
- Critical Reflection: Transformative learning goes hand in hand with critical reflection. It becomes very essential to critically analyze issues at hand.
- Direct and active experience: It is found that the most important aspect is the direct experience that are meaningful to learners
- Readiness for the transformative experience: It is found that individuals awareness and readiness encourages the way towards transformative learning.
- Discourse: It has been found out that discussion is a critical aspect of transformative learning and there are many advantages of doing it online.

- **In Practice** :
- When transformative learning is the goal of adult education, then the environment in which it should be achieved are the following:
- The role of the Educator
- The role of professional development of the Educator
- Strategies for transformative professional development
- Examples in Educator professional development
- The role of learner
- The role of the rational and Affective
- **Examples of Transformative Learning**

- **Examples of Transformative Learning**
- Exposure to different cultural experiences can facilitate the expansion of one's worldview, as well as foster empathy and an appreciation of diversity
- Learning and Development(L&D) leaders should serve as guide and facilitators rather than direct instructors to promote autonomous thinking and growth.
- Enhance personal questioning through journaling as a way of cultivating critical awareness of assumptions.
- Enable learners to play an instrumental role in their own learning. This might be achieved in corporate training through creation of professional learning team.

- According to the UNESCO's Global Action Programme (2014) “transforming learning and training environments” has been considered as one of its five priority action areas.
- Critical reflection about the social context has to be encompassed under Transformative Learning which also includes the educational environment as an institutional setting subject to power relations.
- Transformative learning is not possible in reality without transformative teaching, which should include an emphasis on personal experience, self-organized knowledge, values and emotions; inter-and transdisciplinary (Balsiger, 2015).
- A role shift is also necessary. Teachers take on the role of coaches, facilitating learning and co-learning not only among students but also between students and teachers who are major stakeholders in society.
- The UNECE-ESD Competency Framework (2011) highlights what educators need for teaching sustainability: holistic approaches, enlightening change, and achieving transformation

Implications for Indian Higher Education Policy Research and Practice-

National Knowledge Commission of India (2007) in its report on Higher Education stated that "Universities must become the hub of research once again to capture synergies between teaching and research that enrich each other.

This requires not only policy measures but also changes in resource allocation, reward systems and mindsets."

The Commission also stated that the importance attached to research has eroded steadily over time and the volume of research in terms of frequency of publication and quality of research reflected in the frequency of citation or the place of publication, on balance, is simply not what it is used to be.

Research and Development/R&D programs in India and abroad also can be made effective if it is based on Participatory Research Approach.

Participatory Sustainable Development is a process through which people influence and share control over development initiatives, decisions and resources that affects them. People are able to reflect critically. They are allowed in the decision-making and designing part of the research process. They feel encouraged and empowered as they become partners in decision making and contribute to creative problem solving.

- The objective of the participatory approach is basically for long-term sustenance. It causes both qualitatively and quantitatively changes in the life of the participants. The participatory approach not only brings about changes in the economic indicators of development but also in the social structure and psychological wellbeing of the people (Sudhir & Soundari, 2006).
- **Transformative Learning: A Paradigm shift for Indian Higher Education**
- The New Education Policy (2020) to promote education among the people of India is sure to improve the quality of education being rolled out. The policy covers elementary education to colleges in both rural and urban India. The new policy ensures to cover a wide ambit from early childhood to higher education to professional education to vocational education to teacher education and training to professional education.
- It is based on the ground reality of the country's education scenario that puts more emphasis on the creativity and innovation as well as personality development of the students rather than expecting them to score high and memorizing the content without getting a basic grasp of concepts.
- The NEP looks to improve the quality of education of our system in the following ways:

- **1. Change in the pedagogical structure**
- The 10+2 board examination structure has been dropped; the new school structure will be 5+3+3+4, which comes as a big relief; would prove revolutionary, if implemented in full faith and paper. While most of the existing private schools already have the Elementary Child Care Education/ ECCE embedded in their system and will only have to make a slight change in the class structure and objectives of the change.
- **2. Transforming the Teaching-learning Process**
- a. Focus on core essentials: Mapping of the curriculum across grades and narrowing it to the respective core knowledge only. The focus will be on practical application-based learning. This reduction will create space for teachers to add activities related to experiential learning, creative thinking and critical skills.
- b. Stress on the importance of literacy/numeracy skills: All schools will have to rework in these areas to bring about a transformation in the teaching strategies so that these foundational skills can be developed, strengthened and achieved by Grade 3. There will need to be more focus at an early age on reading, writing, and learning of basic mathematical concepts. Introducing innovative teaching would be essential to achieve this.

- c. Focus on the interest area: A major change is required in the existing educational framework policy. The New Education Policy has tried to remove this flaw by addressing that the interest area of a student needs to be focused upon. And from a young age, they should be given internship opportunities to learn and make advancements in their interested fields in the interest of the nation.
- 3. No separation of subject or field: There will be no hard separation among ‘curricular’, ‘extracurricular’, or ‘co-curricular’, among ‘arts’, ‘humanities’, and ‘sciences’, or between ‘vocational’ or ‘academic’ streams. Subjects such as physical education, the arts and crafts, and vocational skills, in addition to science, humanities, and mathematics, will be incorporated throughout the school curriculum.
- 4. HECI- Common regulatory body for entire Higher education:
- Higher Education Commission of India (HECI) will be established as a single overarching umbrella body the for whole higher education, excluding medical and legal pedagogy. HECI to have four dominant verticals - National Higher Education Regulatory Council (NHERC) for regulation, General Education Council (GEC) for standard setting, Higher Education Grants Council (HEGC) for funding, and National Accreditation Council(NAC) for accreditation.

- 5. Content will focus on idea, application and problem- solving: The instructed content will concentrate on key concepts, ideas, applications, and problem-solving. Teaching and learning will be administered in a more interactive way.
- 6. National Research Foundation (NRF) to be established: A National Research Foundation (NRF) will be ascertained. The overarching goal of the NRF will be to facilitate a culture of research to penetrate through universities. The NRF will be governed, independently of the government, by a rotating Board of Governors comprising of the very best researchers and innovators across fields.
- The new National Education Policy (NEP,2020) has indicated an obvious will to move the needle away from the obsolete world of learning. This has been accentuated with the triad of multidisciplinary higher education, multiple options at senior school and multiple chances of accomplishment in school-leaving examinations. The emphasis on foundational learning, the inclusion of the very young into formal learning and the emphasis on holistic learning are goals that are baked into the policy. As institutions implement these changes, we will see a transition in the quality education system to one where students receive more conceptual and practical education.

- **Pedagogical Framework of Transformative Learning**
- Transformative learning has materialized within the domain of adult education as an influential image for understanding how adults learn. It has attracted researchers and practitioners from a wide variety of theoretical persuasions and practice settings, yet it is a sophisticated idea that offers substantial theoretical, practical, and ethical challenges. Service-learning environments are complex learning contexts that generate a level of disequilibrium or anxiety that may or may not result in transformative learning. (Hinck and Burton, 2015). The importance of faculty development and training as a means to prepare faculty to design the capstone course as a high-impact educational practice should be emphasized upon. (Martin and Strawser, 2017). "Transformative pedagogy" shares some important aspects with other constructs or design frameworks. For example, critical pedagogy may be characterized as an ideology of education that emerged from a legacy of radical social thought and progressive educational movements that inspired the linking of schooling to democratic principles of society, and to transformative social action in the interest of oppressed communities (Sen, in The Critical Pedagogic Reader, 2008, p.3).

NPTEL

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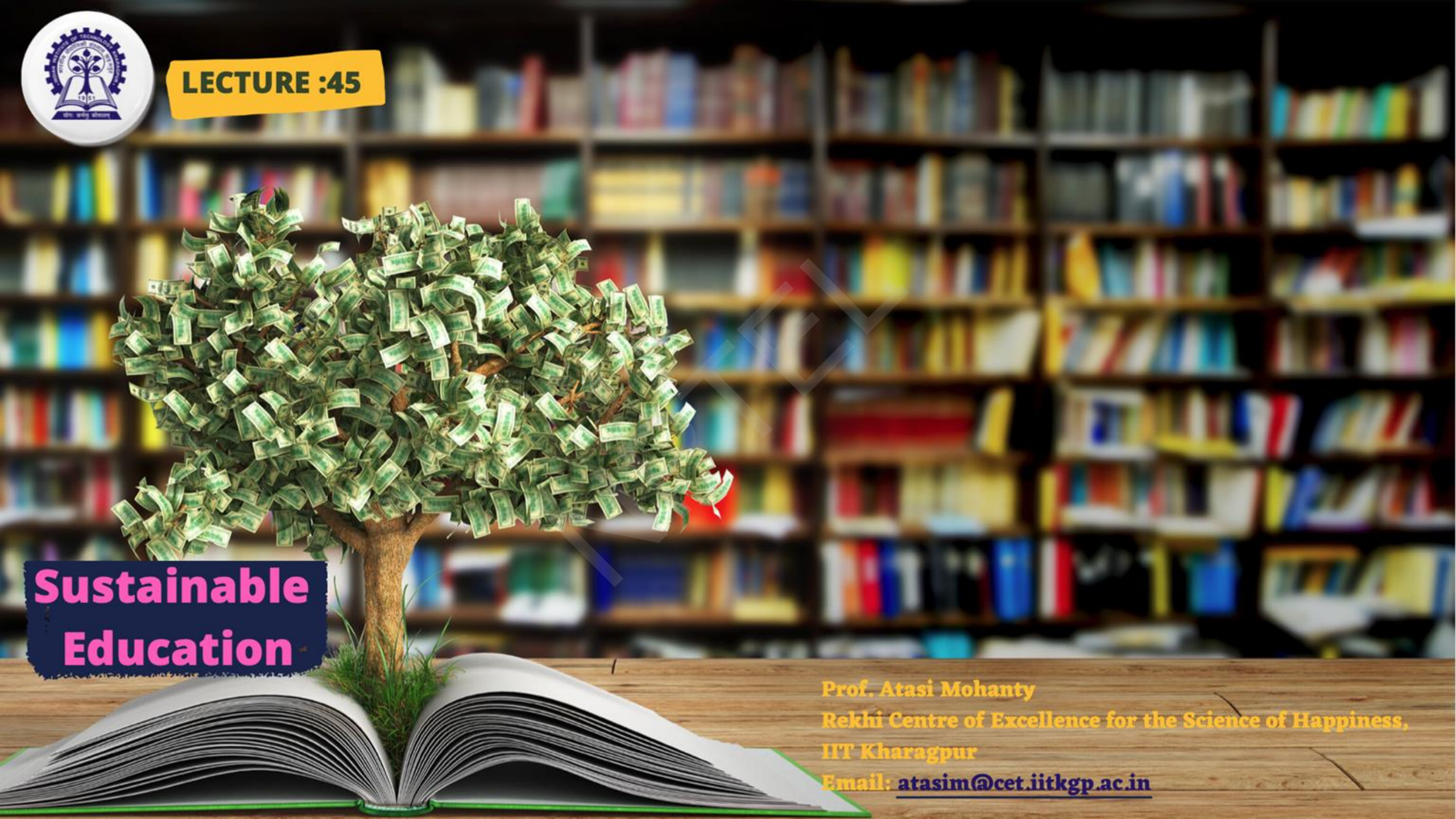

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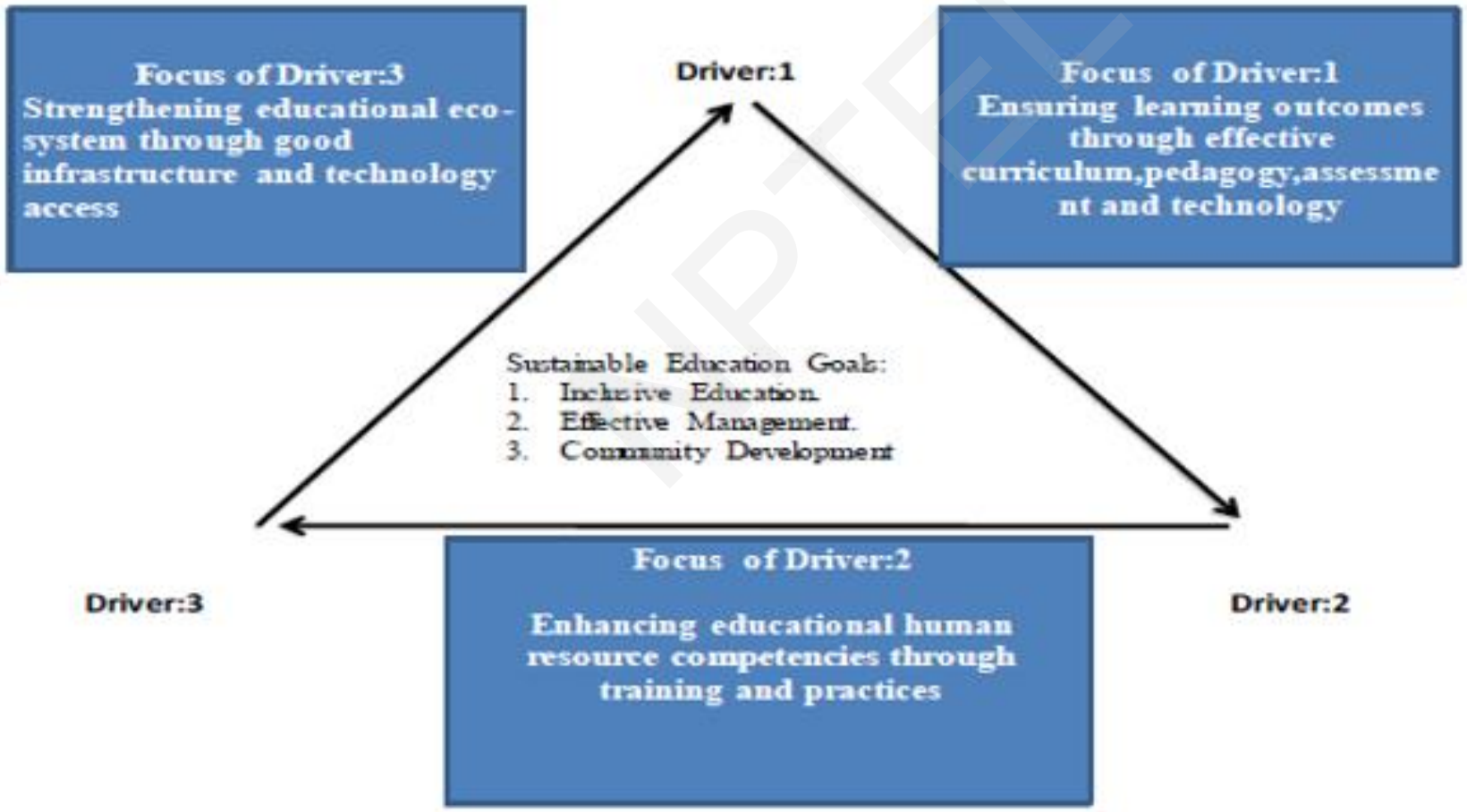
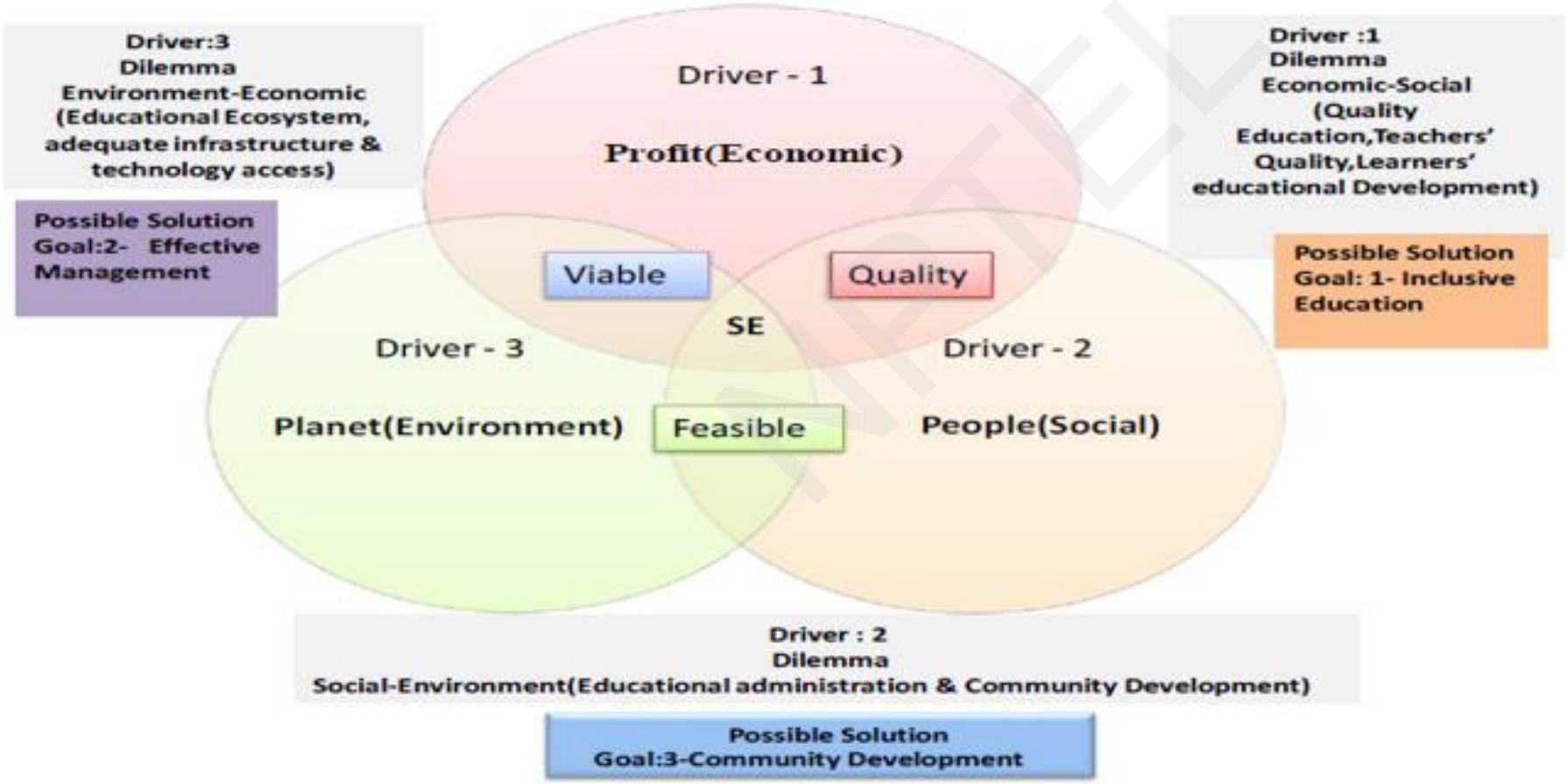
- **Developing a Conceptual Model of Transformative Learning for the Indian Education system-**

- **A Conceptual Model of Sustainable Education for India**

A conceptual sustainable education triangle-

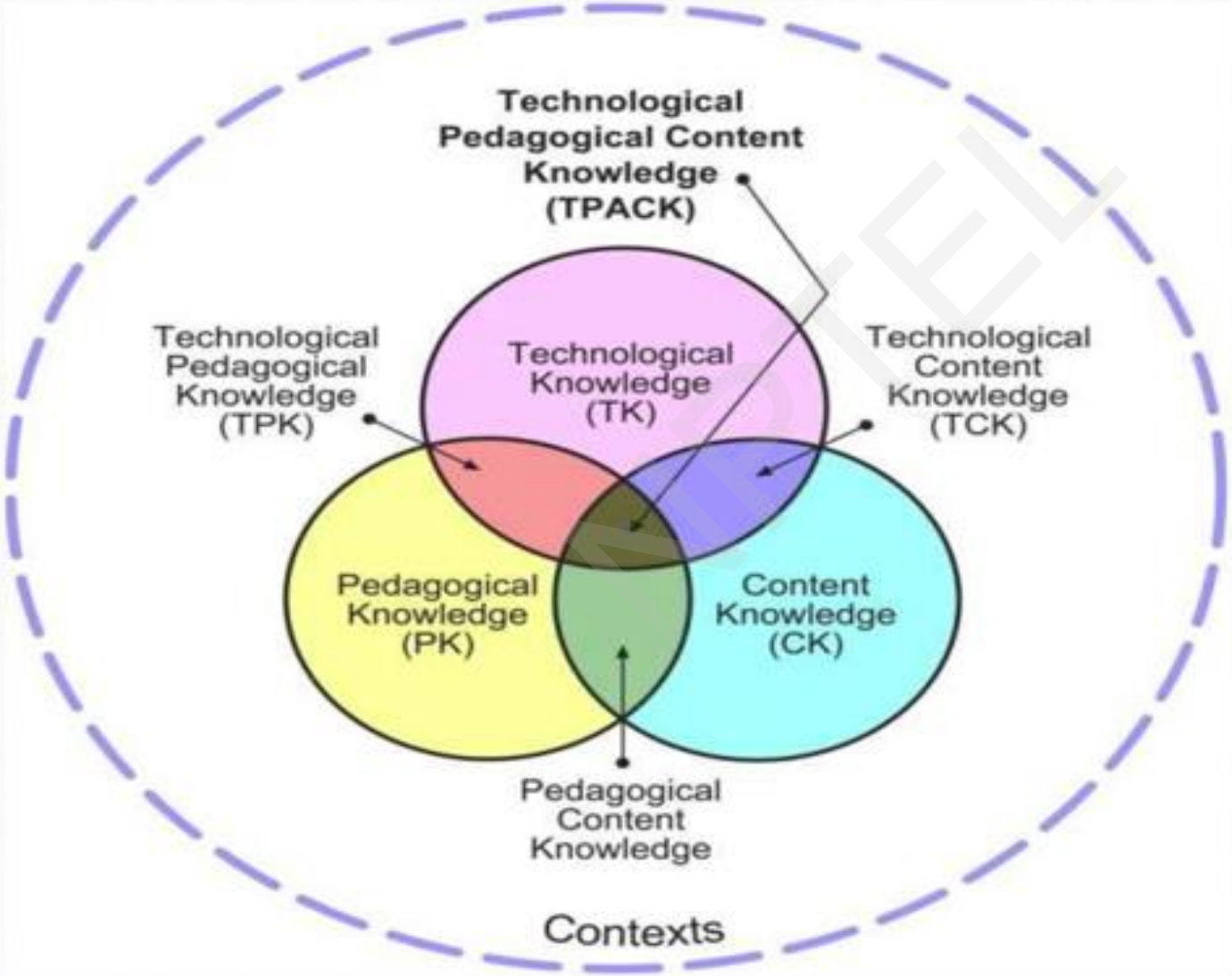


A conceptual dilemma model of sustainable education-



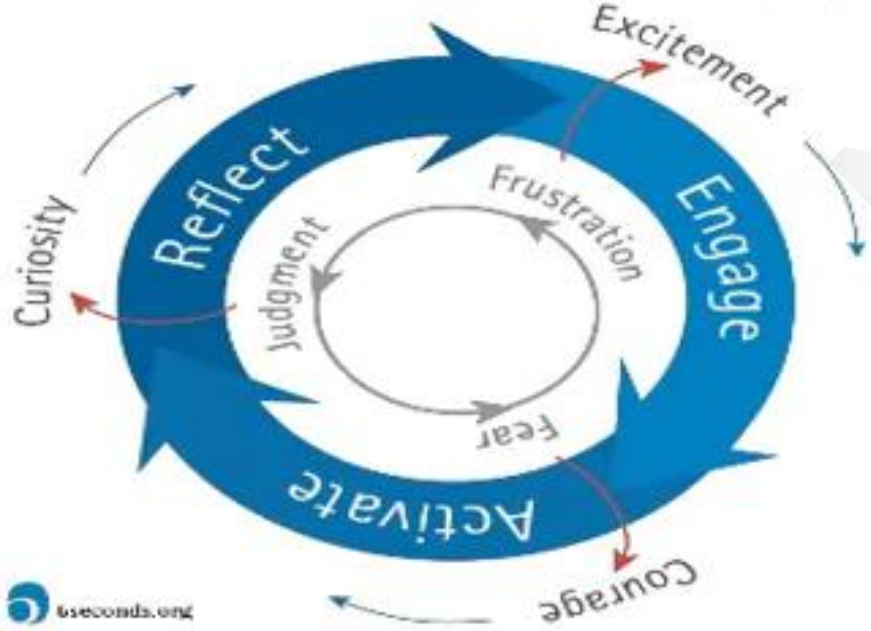
- According to (Mohanty,2019) ‘Sustainable Education’ model the vital components are, i.e., the three drivers – Driver1-profit, Driver 2 – people, Driver 3 – the planet would be like- the 1st component Driver – 1 is the ‘Ministry of Human Resource Development’ (MHRD) of Govt. of India as the major policymaker, decision-taking body and economic resource provider for running a centralized board of education in the country; hence called the ‘Profit’.
- The 2nd Driver is ‘people’ engaged in the education sector, the human resources (all direct and indirect stakeholders) of our school education system (primary or secondary level of school education);
- the 3rd Driver is ‘planet’ the education set up, school environment or can be called the “educational ecosystem” to carry on all the teaching – memorizing activities and educational administrations / transactions.

- All these three drivers (profit, people and planet) are expected to work together harmoniously to make the education system a sustainable one. Thus, the objective of Driver 1 (profit) would be – “Ensuring learning outcomes through effective curriculum, pedagogy, assessment and technology”.
- The objective/ focus of Driver 2 – (people) would be – “Enhancing educational human resource competencies through training and practices”.
- The objective/ focus of Driver 3 – (planet) would be – “Strengthening the educational eco-system through good infrastructure and technology access. New challenges to learning are emerging and 21st century education must address to these and contribute to greater humanity in a rapidly changing world (UNESCO, 2015).
- Education has been called upon to support this transformation (WBGU 2011). The 2005 to 2014 ‘U.N. Decade of Education for Sustainable Development’ (DESD) confirmed that ESD would enable us “to constructively and creatively address present and future global challenges and create more sustainable and resilient societies ”(UNESCO 2017).
- ESD needs to be built on transformative learning and explicitly inculcate critical reflection on values. . At the same time, transformative learning should not be used to regulate learners but to empower them for autonomous critical action.


Technological Pedagogical Content Knowledge (TPACK)
Technological Pedagogical Knowledge (TPK)
Technological Knowledge (TK)
Technological Content Knowledge (TCK)
Pedagogical Knowledge (PK)
Content Knowledge (CK)
Pedagogical Content Knowledge
Contexts

transformative LEARNING

People may not always deliberately set out to critically question their beliefs and values; many times transformative learning is prompted by an outside event and that event may be unexpected, hurtful, or devastating (Cranton, 2006, p. 6)



Transformative learning is a process of examining, questioning, validating, and revising our perspectives (Cranton, 2006, p. 23)

21ST CENTURY
eLEARNING
DIGITAL EDUCATION
CONNECTIVISM
SMART LEARNERS
TRANSFORMATIONAL LEARNING
NEW THINGS IN NEW WAYS
EVALUATION
OLD THINGS IN NEW WAYS
OLD THINGS IN OLD WAYS
INTEGRATION PROCESS
START HERE
efficacy
creativity
mobility
collaboration
differentiation
INTEGRATION ELEMENTS
VISION
Align process with strategic vision for physical & digital spaces
LEADERSHIP
Leadership to develop workforce
Integrated with curriculum
Policy change as necessary
COMMUNITY
Parents & wider community involved
Cultural elements integrated as strengths
School culture of innovation

- **Conclusion**

- In the recent years, sustainability has become a slogan for raising social, economic and environmental awareness. Higher Education Institutions (HEIs) play an important role in promoting development principles and practices. In addition to educating students on sustainability topics, these institutions should take on a leadership role in incorporating sustainable practices into their services and operations and in transforming their local and regional communities into more sustainable communities. Acknowledging the urgent need for Sustainable development and the importance of research in this process, universities and their researchers bear the fundamental and moral responsibility to contribute with their research to Sustainable development.

- **The Role of Institutions**
- The presence of a new international agreement or a new legislation alone is not enough to guarantee action.
- The SDGs' ambitious vision must be converted into specific action plans and must find a foothold in existing institutions of governance to have impact. It takes time for global and regional institutions to change; it takes longer for all national policypriorities,legislations,planningcycleinstitutionalarrangements,programmes, and modes of working to be dovetailed with the new paradigm, and for human, financial, and technical capacities to be put in place.
- For implementation to happen in an effective manner , governments must invest resources and time for:

- Awareness raising, both among citizens at large, but also more critically among those tasked with the Agenda's implementation.
- The need for strengthening awareness of the specifics of the Agenda among policy makers, senior officials, and front-line workers responsible for policy implementation.
- Developing the ability to exercise the right to education through modifying existing national policies, legislations , and plans in line with new SDG commitments, allotting funds necessary for their implementation, strengthening implementation mechanisms, and strengthening monitoring systems.
- Strengthening their ability to enforce the right to education through strengthening existing redress and state accountability mechanisms.
- Building a wider community of SDG supporters that can push for the Agenda's implementation by creating enabling environments for civil society and citizen participation in the governance processes.

- **References**

- https://www.youtube.com/watch?v=j8aXSTN71MY
- https://en.unesco.org/news/global-media-and-information-literacy-week-media-and-information-literacy-public-good
- file:///C:/Users/Prof.%20Atasi%20Mohanty/Desktop/ESD%20&%20SDG%204-2022/SDG4%E2%80%99s%2010%20targets%20_%20Global%20Campaign%20For%20Education.html
- https://www.ukfiet.org/2020/post-covid-19-and-sdg4-window-of-opportunity-or-of-opportunism/
- https://en.unesco.org/news/united-nations-alert-education-should-be-clear-priority